Incorporating Duration in Regime-Switching Strategies: A Comparative Study of HMM vs HSMM

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Abstract

This paper tests whether explicit duration modelling improves regime-based allocation between two equity strategies: a momentum sleeve and a mean reversion sleeve. Using daily S&P 500 price-based features, I estimate a two-state Gaussian Hidden Markov Model (HMM) and a two-state Hidden Semi Markov Model (HSMM) with identical emissions, retraining scheme, and hard gating rule. The HSMM constrains regime durations with shifted Poisson sojourn distributions, whereas the HMM implies geometric durations. Both models switch between the sleeves in an expanding-window out-of-sample backtest from 2006 to 2024. I compare performance, regime characteristics, and behaviour in algorithmically detected persistent drawdowns and volatility spikes. Both gated strategies increase Sharpe ratios relative to the underlying sleeves. The HSMM produces more distinct return regimes and sometimes maintains defensive positioning longer in drawdowns, but tail risk reductions are modest and episode dependent. Overall, the HSMM and HMM gated portfolios perform very similarly, with the HSMM delivering slightly lower maximum drawdown and CVaR at the cost of slightly lower returns. Accounting for transaction costs widens the HMM's advantage, because the HSMM spends more days in the high-turnover mean reversion sleeve.

Highlights:

- Compare HMM and HSMM regime detectors gating between momentum and mean reversion sleeves, holding everything else constant.
- Both gated strategies raise the Sharpe ratio to about 0.9, though not significantly at the 5% level; HSMM has minimal reductions in drawdowns and CVaR versus HMM.
- HSMM stays defensive longer in crises, reshaping the path of risk rather than average performance.

1. Introduction

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Financial markets often exhibit distinct regimes of behaviour, such as bull vs. bear markets or high- vs. low-volatility periods. Identifying these regimes in real time can enable investors to adjust portfolios and mitigate downside risk during unfavourable conditions. A popular approach for regime identification is the use of hidden Markov models (HMMs), which infer an unobserved state (the market regime) from observable data like returns and volatility. HMM-based strategies have been shown to improve risk-adjusted performance by switching asset allocations according to the inferred regime (Bulla et al., 2011; Nystrup et al., 2015). However, a limitation of standard HMMs is their implicit assumption that regime durations follow a geometric distribution, meaning regimes have a memoryless property. In practice, market regimes (especially bear markets or crises) often exhibit duration dependence, for example, a prolonged bear market may be more likely to continue the longer it has lasted, up to a point (Durland & McCurdy, 1994). This insight has led to extensions of HMMs that incorporate duration dependence, such as hidden semi-Markov models (HSMMs), which allow explicit specification of the state sojourn time distribution (Bulla & Bulla, 2006).

In this study, I investigate whether incorporating an explicit duration component into a regime-switching model improves the performance of a tactical asset allocation strategy. My focus is a strategy that switches between a momentum portfolio and a mean-reversion (defensive) portfolio, depending on the predicted market regime. Momentum strategies are known to perform well during sustained upward trends but can suffer abrupt losses during regime shifts or market crashes (Daniel & Moskowitz, 2016). Mean-reversion or defensive strategies, on the other hand, tend to perform better during choppy or declining markets by capitalizing on market overreactions or by reducing exposure. By dynamically allocating between these two strategy sleeves, one can potentially capture upside during favourable regimes and avoid large drawdowns during adverse regimes (Bulla et al., 2011). The key question I address is: Does a regime identification model that accounts for state duration (HSMM) provide a significant advantage over a standard HMM (with memoryless state durations) in guiding such a switching strategy?

My hypothesis is that the HSMM, by recognizing how long a regime has persisted, will better differentiate between transient corrections and persistent bear markets. In theory, this could enable the strategy to stay with the momentum sleeve during brief, transient pullbacks (avoiding premature switching), yet switch more decisively to the defensive sleeve during prolonged downturns. The standard HMM, lacking an explicit duration awareness, might either overreact to short-lived drops or underreact in early stages of a longer crash due to its geometric duration bias. On the other hand, during short, irregular drawdowns, duration dependence may hinder the HSMM and make it less versatile. I test this hypothesis on historical data for the S&P 500 (SPY) over a large time frame. I construct and compare the two regime-switching models, one HMM and one HSMM, that are identical everywhere (feature set, number of states, etc.) except for the incorporation of duration dependence in the HSMM. Both models will then be used to allocate between the momentum and mean-reversion sleeves. I then evaluate the out-of-sample performance of the resulting strategies in terms of returns, volatility, drawdowns, and risk-adjusted measures.

The paper is organized as follows; in the literature review, I discuss past research on regimes in financial markets, allocation strategies based on these regimes, and HMM/HSMM models. The methodology section describes my data, the momentum and mean-reversion strategy sleeves, and the construction of the HMM and HSMM model. The results of my empirical analysis will then be presented, including overall performance comparisons and a closer look at how each model

behaved during significant market drawdown episodes. Then I will conclude, highlighting the findings, acknowledging limitations, and suggesting directions for future research.

2. Literature Review

Early work on regime-switching in finance can be traced back to Hamilton (1989), who introduced a Markov-switching model to capture shifts in the business cycle. Since then, Markov regime-switching models have been widely applied to characterize bull vs. bear markets and high vs. low volatility regimes in equities (Guidolin, 2011). These regime-switching models assume that market returns are driven by an unobserved state that follows a Markov chain. This framework has proven useful to explain volatility clustering and non-linear return dynamics. For example, Ang and Bekaert (2002) showed that accounting for regime shifts can improve international asset allocation decisions, as optimal portfolios shift towards safer assets in "bad" regimes. Similarly, studies have found that switching to strategies which reduce equity exposure during volatile, or bear regimes can significantly lower drawdowns and improve Sharpe ratios (van Vliet & Blitz, 2011; Bulla et al., 2011).

The hidden Markov model (HMM) has been used in numerous financial applications to detect these Markov chains. In an HMM, the regime is inferred from observable data (e.g., returns, volatility) using estimated state-dependent emission distributions. Researchers have used HMMs for tasks such as market timing, volatility forecasting, and strategy switching. Rydén et al. (1998) demonstrated that a simple two-state Gaussian HMM can capture many of the properties of daily stock index returns, including volatility shifts. Subsequent works found that HMM strategies can enhance performance by predicting regime changes: for example, Bulla et al. (2011) applied a two-state HMM to switch between risky and safe assets, finding that certain HMM-based allocation rules outperformed static allocations. Nystrup et al. (2015) achieved improved returns from a long-short equity strategy using HMM regime signals. HMM regimes have generally also been able to identify intuitive market conditions (e.g., a "calm" state with modest positive returns and low volatility, and a "turbulent" state with high volatility and potential negative returns), making them interpretable and useful for risk management (Ang & Timmermann, 2012).

One well-documented vulnerability of momentum strategies is the occurrence of momentum crashes that tend to happen when market regimes abruptly change from a steady upward trend to a sharp reversal (Daniel & Moskowitz, 2016). The 2009 rebound after the financial crisis, for example, saw prior losers gain (as value stocks recovered) while the momentum portfolio (long winners) suffered large losses. To adapt to this, researchers have explored dynamic momentum strategies that time the market's state. Daniel, Jagannathan, and Kim (2019) developed a two-state HMM (turbulent vs. calm) specifically to time momentum exposure. In their model, the momentum strategy is only fully applied in the calm state and is scaled down in the turbulent state to avoid crashes. They reported that this HMM-based strategy avoided major crash events and had superior risk-adjusted returns out-of-sample (Daniel et al., 2019).

Despite the success of HMMs, they have known limitations. One key issue is the geometric sojourn time inherent in a first-order Markov chain. In a standard HMM, the probability of remaining in a state decays exponentially with time spent in that state:

If a_{ii} is the self-transition probability for state i, then the probability that the state lasts for d consecutive periods is:

$$P(D = d) = (a_{ii})^{(d-1)}(1 - a_{ii})$$

This geometric distribution has a constant hazard rate, so the probability of exiting the state is independent of how long it has already lasted. The geometric distribution may not accurately represent the duration of financial regimes. Bear markets tend to exhibit duration dependence, for example, very short bear phases and very long ones are less common than intermediate lengths, and so the hazard of exiting a bear market can change over time (Pagan & Sossounov, 2003). Bulla and Bulla (2006) argued that standard HMMs cannot reproduce such properties, so they introduced the use of hidden semi-Markov models (HSMMs) in finance, which allow the sojourn time distribution (the distribution of state durations) to be specified rather than being forced to geometric. By using an HSMM or other duration-dependent approaches, one can impose, for example, a minimum regime length or a more heavy-tailed duration distribution that better matches observed regime lengths (Durland & McCurdy, 1994; Bulla & Bulla, 2006).

The idea of duration-dependent Markov switching was formally explored by Durland and McCurdy (1994) in the context of business cycles. They extended Hamilton's model by making transition probabilities a function of the time already spent in the current state, effectively altering the likelihood of switching as a state grows older. There have been relatively few applications of HSMMs in finance, but recent work by Baitinger and Hoch (2024) provides a direct comparison between HMM and HSMM for regime-based allocation. They found that while an HSMM (with Poisson or Gamma-distributed state durations) fit the data better in-sample, it did not significantly outperform a simpler HMM in out-of-sample investment performance. This suggests that the added complexity of modelling duration might not be worth it, a point I aim to further investigate.

In summary, the literature indicates that regime-switching strategies can add value by adapting to market conditions, as shown by numerous HMM-based approaches. However, the question of how to best model regime duration remains open. Traditional HMMs assume a geometric duration, which might be suboptimal if true regime lengths have different distributions. HSMMs offer a more flexible alternative, but prior studies (Bulla & Bulla, 2006; Baitinger & Hoch, 2024) provide mixed evidence on whether this complexity actually improves investment outcomes. My work contributes to this literature by specifically examining a momentum vs. mean-reversion switching strategy, directly comparing an HMM and HSMM. I use realistic strategy performance metrics to evaluate whether duration modelling helps the strategy distinguish between short, transient drawdowns and longer, more dangerous crises. It extends the out-of-sample comparison of Baitinger and Hoch (2024) from regime-conditional portfolio weights to hard switching between two strategy sleeves with very different payoff profiles.

3. Methodology

Data and Base Strategies

My analysis uses historical daily OHLC data for the S&P 500, using the SPDR S&P 500 ETF (SPY) as a proxy for the market. The sample spans multiple market cycles (I use data from 2002 through 2024, covering major downturns such as the 2008–09 Global Financial Crisis and the 2020 COVID crash). I derive 5 features that include both return-based and volatility-based measures commonly used to detect regime shifts:

Recent return indicators: 1-day return and 20-day return.

Volatility measures: Parkinson's 20-day realised volatility, body to range ratio, and 10-day sign flip rate

These features collectively help the models discern patterns characteristic of regimes, for example, a combination of low volatility and positive momentum might indicate a bull trend, whereas high volatility and negative short-term returns might indicate a correction or bear phase.

I design two baseline strategy sleeves that the regime model will toggle between. Both are built from CRSP daily returns of S&P 500 constituents using point-in-time index membership, so the stock universe is free of survivorship bias:

Momentum sleeve: this strategy is meant to thrive in trending upward markets. In my implementation, the momentum sleeve is a trend-following strategy that simply buys the top 50 performers in the S&P 500 based on a combination of their 12-, 6-, and 3-month momentum. The strategy stays fully invested in the market and rebalances weekly.

Mean-reversion (defensive) sleeve: the other strategy is built to perform well in volatile or declining markets by reducing market exposure and taking contrarian positions. My mean-reversion sleeve is once again kept simple by selling the top 50 and buying the bottom 50 performers in the S&P 500 based on 5-day returns. A new long-short portfolio is formed every day and held for three trading days, so the sleeve always holds three overlapping portfolios. The strategy has close to no net market exposure.

These two sleeves are constructed to have very different if not opposite return profiles in regimes. Exhibit 1 compares the annualized historical performance of the momentum sleeve vs. the mean-reversion sleeve over the period 2006 - 2024.

Exhibit 1: Performance comparison between momentum and mean-reversion sleeve (2006 - 2024)

Sleeve	Return	Volatility	Sharpe	Max Drawdown	CVaR 95%
Momentum	9.8%	20.8%	0.47	-53.2%	3.2%
Mean - Reversion	14.7%	21.4%	0.69	-49.5%	2.9%

Surprisingly the mean-reversion sleeve shows much higher average returns with only a little higher volatility. This gives the mean-reversion sleeve a much greater Sharpe ratio (risk-free rate as 0) and overall performance. The maximum drawdown for both strategies is extensive at around 50% aligning with previous research and observations that state the importance of regime identification. While the sleeve performance goes against the intuition that momentum would have superior performance, this does not have an impact on the research. My goal is to compare performance in using regime identification to allocate between these sleeves, trying to capture the best of both: stay with momentum (risk-on) during favourable regimes and switch to defensive (risk-off) during high volatility regimes.

Regime Modelling with HMM and HSMM

I consider a two-state hidden Markov model (HMM) as my primary regime identification tool. The choice of two states is guided by interpretability (roughly corresponding to "calm/uptrend" vs "volatile/downtrend" regimes) and is supported by prior work that finds that two or three regimes often suffice for univariate equity series (Guidolin & Timmermann, 2006). I label the states for convenience as 'Trend' and 'Chop'.

HMM Specification: I implement the set of models from hmmlearn in python to create this model. As required for the HMM, I assume that the observed data at time *t* (the feature vector including returns and volatility measures) is generated from one of two multivariate Gaussian distributions, depending on the hidden state $S_t \in \{0,1\}$. Both states have means and covariances of the features as parameters for the Gaussian emissions. The state evolves according to a Markov chain with transition probabilities in the form of a matrix A where the diagonals are defined as:

$$P(S_t = i | S_{t-1} = i) = a_{ii}$$

where S_t denotes the state at time *t* and a_{ii} can be interpreted as the respective probabilities of staying in the same state (regime). The probability of switching or transitioning to the other state (the off-diagonal elements) is thus defined as:

$$P(S_t = j | S_{t-1} = i) = 1 - a_{ii} = a_{ij}$$

where state $i \neq j$. The elements a_{ii} and a_{ij} (probability of staying and leaving the state) logically add up to 1.

These parameters are estimated via maximum likelihood, using the Expectation-Maximization algorithm on the entire training dataset. The fitted HMM creates a filtered probability each day for being in state 0 or 1. A critical property of the standard HMM is that the distribution of regime durations is geometric or 'memoryless' as previously discussed. This property implies that no matter how long the model has been in the bear state, the probability of the regime ending the next day is always constant. This geometric assumption could lead the HMM to misrepresent the probability of regime durations.

HSMM Specification: In the hidden semi-Markov model (HSMM), I modify the regime process to model how long the market remains in each state. As in the HMM, I assume that returns and volatility features are generated by a hidden two-state process. Conditional on the state, the observations follow a multivariate Gaussian distribution with state-dependent mean and covariance. The difference is that I no longer let regime lengths be implied by self-transition probabilities in a Markov chain (the geometric distribution). Instead, I introduce a separate sojourn time distribution for each state.

In this two-state application it is easiest to describe the regime process in terms of segments. When the model enters state *i*, it first draws a duration D_i from a state-specific sojourn distribution. The process then remains in state *i* for exactly D_i consecutive trading days. Once this segment ends, the regime switches to the other state, and a new duration is drawn. In terms of implementation, this alternation is enforced in the mhsmm package by specifying an embedded transition matrix with zero diagonal entries and off-diagonal entries equal to one. As a result, all persistence in regimes comes from the sojourn distributions and from the choice of initial state, rather than from a non-zero probability of staying in the same state as with the HMM.

I model the sojourn time D_i in state i as a shifted Poisson random variable,

$$D_i = shift_i + Y_i\,, \quad Y_i \sim Poisson(\lambda_i)$$

The corresponding probability mass function for the duration in state i is:

$$P(D_i = d) = \frac{e^{-\lambda_i}\lambda_i^{d-shift_i}}{(d - shift_i)!}, \quad d \geq shift_i$$

This specification allows us to approximately encode that some regime lengths are more probable than others. In contrast to the geometric sojourn time in the HMM, the exit probability from state i now depends on how long the process has already been in that state. The hazard of exiting a regime is therefore duration dependent rather than constant.

As stated above, I implement the HSMM using the mhsmm package in R, specifying a two-state Gaussian HSMM with shifted Poisson sojourn distributions. As with the HMM, the model is estimated by maximum likelihood using an EM algorithm for HSMMs and produces smoothed and filtered probabilities of being in each state over time.

Training and Backtest Design

To analyse how the HMM and HSMM perform in guiding allocation decisions, I embed both models in an equal backtesting framework. The aim is to compare how each model behaves out-of-sample when it is used to switch between the momentum and mean-reversion sleeves. To isolate the impact of duration modelling, I ensure that every aspect of the setup is the same for the two models except for the duration process itself.

The 5 SPY features are first aligned with the strategy return data. I start the sample in 2002 and reserve an initial period for model training only. The first out-of-sample allocation decision is made in 2006, after several years of data have been used to train the models. From that point onward, both models operate in an expanding window setting.

The backtest proceeds as follows:

1. At the first allocation date, I fit the model (HMM or HSMM) on the full history of the 5-feature data up to the previous day. The features are standardized within the training sample by subtracting the sample mean and dividing by the sample standard deviation for each feature (z-scored). The model parameters are thus estimated on this scaled data.

2. At the start of each trading day in the out-of-sample period, I use the filtered probabilities up to the prior day to form a regime prediction. The allocation rule with this prediction is then simple; I allocate to the sleeve whose corresponding regime is predicted as most likely.

3. At the end of the day, the realised feature vector for SPY and the strategy return are observed. The model updates its filtered probabilities given the new observation, and the process repeats.

4. Every 21 trading days (approximately once per month), the model parameters are re-estimated on an expanding window that incorporates all data up to the day before retraining. The rescaled features, estimated state-dependent means and covariances, and regime labels are updated accordingly.

This procedure is applied identically for the HMM and HSMM, the underlying sleeves, feature engineering, retraining schedule, and allocation rule are the same in both cases.

4. Results

In this section, I summarize the performance of the HMM and HSMM in my backtest. I first examine the regimes identified by each model in terms of their return and volatility characteristics and effective durations. I then compare the out-of-sample performance of the resulting gated portfolios to the underlying sleeves. Finally, I zoom in on specific episodes of persistent drawdowns and short-lived volatile spikes to see how duration modelling affects defensive behaviour.

Regime Characteristics

I begin by examining how the two models classify regimes. For each model, I group days into trend and chop states based on the regime labels described above and compute statistics for SPY returns in each state.

Exhibit 2: SPY return characteristics by regime for HMM and HSMM

Model	Regime	Ann. Return	Ann. Volatility	Sharpe	Mean length	Median Length
HMM	Trend	8.2%	12.1%	0.68	84.5	43
	Chop	8.2%	29.2%	0.28	42.4	24
HSMM	Trend	10.5%	12.4%	0.84	44.5	13.5
	Chop	4.5%	27.1%	0.17	27.9	7

For the HMM, exhibit 2 shows that the two states differ in volatility rather than in average returns. The estimated trend state has a strong average return and relatively low volatility compared to the chop state which has an equally strong return but much higher volatility. In exhibit 3, the

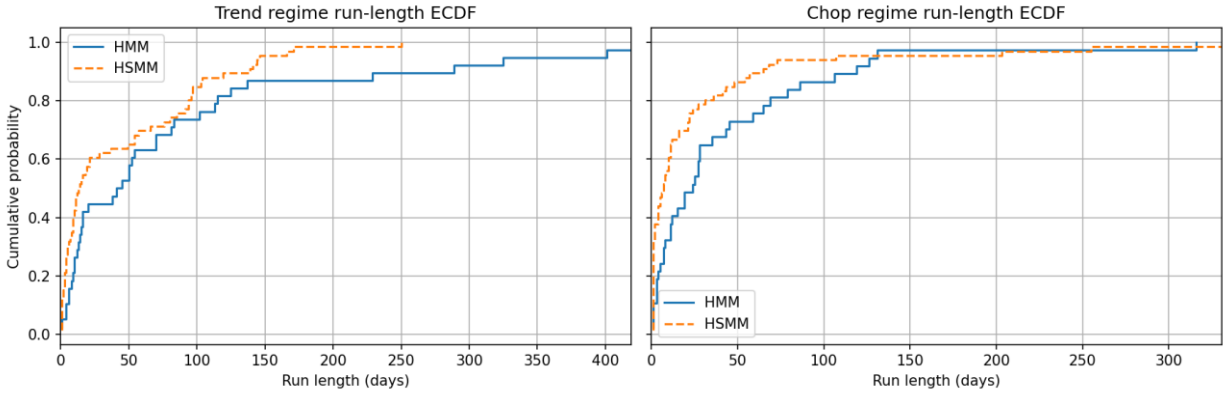
empirical transition matrix of the realised HMM regimes has diagonal entries (probability of staying in the same state) close to one, implying strong persistence or 'stickiness' in regimes. The expected duration of the low volatility state is about 80 to 90 trading days, while the expected duration of the high volatility state is about 40 days.

Exhibit 3: Implied transition matrix for the realised regimes for HMM and HSMM

From state	**To state**			
	HMM		**HSMM**	
	Trend	Chop	Trend	Chop
Trend	98.8%	1.2%	97.8%	2.2%
Chop	2.4%	97.6%	3.5%	96.5%

The HSMM produces a different regime structure. The two states differ both in volatility and in mean return. The trend state exhibits a higher daily average return and low volatility, leading to a higher Sharpe ratio than either state in the HMM. The chop state has both lower average return and higher volatility. In other words, the HSMM tends to identify a truly favourable state and a truly unfavourable state rather than only separating volatility levels. The mean run length of the HSMM trend state is around one to two months, and the mean run length of the chop state is even shorter. Previously I mentioned that the diagonal elements in the embedded transition matrix for the HSMM are 0. This remains true, exhibit 3 instead reports the empirical one-step transition probabilities from the daily state sequence. In this *implied* transition matrix, we once again see a high probability of staying in a given regime in exhibit 3.

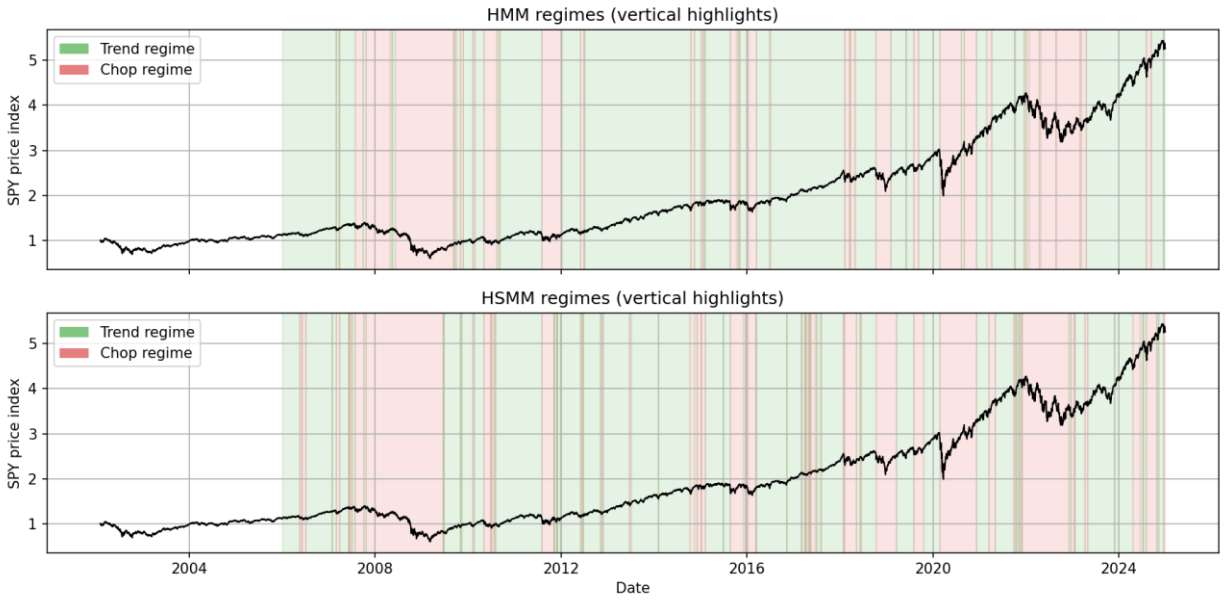
Exhibit 4: Distributions of contiguous regime lengths for the HMM and HSMM



A useful way to visualize these differences is through run length distributions. Exhibit 4 shows that HMM regimes surprisingly have heavier tails in this dataset, in line with the slightly

higher self-transition probabilities. HSMM regimes are shorter, especially in the trend regime where I see a strong difference in tails. Although the original motivation for HSMMs was to better match persistent durations, in my application the HSMM ends up creating shorter but more sharply separated regimes, while the HMM uses a few long runs to capture long distinct phases.

Exhibit 5: SPY price with regime highlights



In exhibit 5, we see that both models are able to capture the main crisis episodes and quiet periods in a similar way. The resulting regime paths are very close, although the HSMM tends to switch between states more often. For example, the relatively calm market environment from 2013 to 2015 is classified almost entirely as Trend by the HMM, while the HSMM inserts several short Chop segments, which is consistent with its shorter implied trend durations.

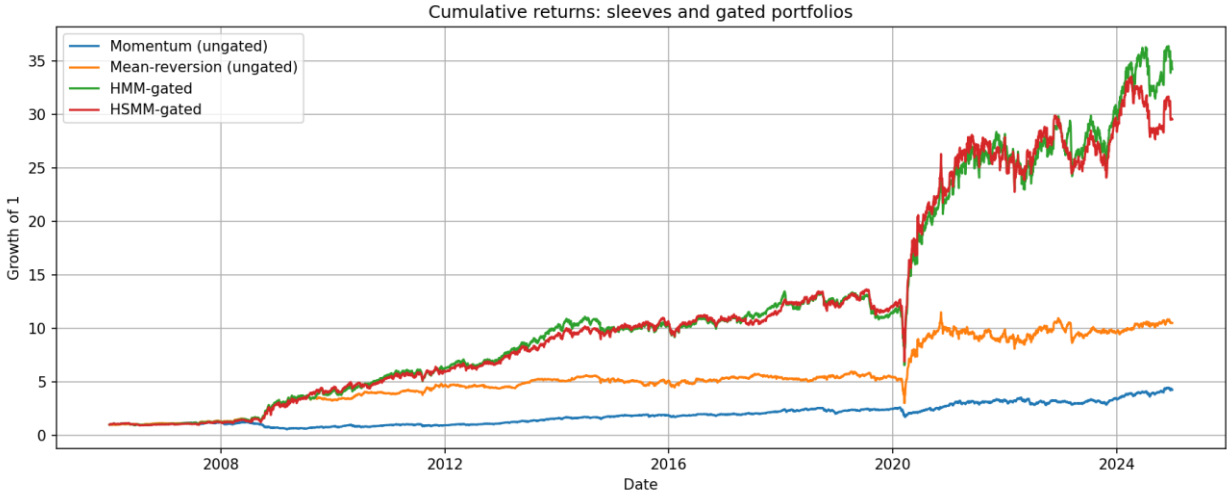
Overall Strategy Performance

I next compare the overall performance of the gated strategies and their underlying sleeves on the days they were allocated to. For each strategy, I compute the annualized return, annualized volatility, Sharpe ratio, maximum drawdown and daily 95 percent CVaR over the full out-of-sample period.

Exhibit 6: Gated and sleeve out-of-sample performance

Strategy	Nr. of Days	Ann. Return	Ann. Volatility	Sharpe	Max Drawdown	CVaR 95%
HMM-gated	4780	21.4%	23.4%	0.91	-51.3%	3.3%
Momentum	3211	10.4%	16.4%	0.63	-28.5%	2.5%
Mean-reversion	1569	43.9%	33.3%	1.32	-48.8%	4.6%
HSMM-gated	4780	20.5%	23.2%	0.89	-49.5%	3.2%
Momentum	2938	11.2%	16.8%	0.67	-21.9%	2.6%
Mean-reversion	1842	35.4%	30.8%	1.15	-47.4%	4.2%

Exhibit 7: Cumulative returns of gated strategies and sleeves



Both gated strategies achieve very similar, strong performance with annualized returns above 20 percent and volatility around 23 percent. The resulting Sharpe ratios are around 0.9, well above either sleeve alone (0.47 and 0.69). A stationary block bootstrap of the Sharpe ratio differences (10,000 resamples, mean block length 21 days) places this improvement at the edge of

conventional significance: p = 0.05 relative to momentum and p = 0.09 relative to mean reversion for the HMM-gated strategy, and p = 0.07 and p = 0.15 for the HSMM-gated strategy. The maximum drawdowns remain large, just over 50 percent for the HMM and slightly below 50 percent for the HSMM. Daily CVaR at 95 percent is in the low 3 percent range for both gated strategies.

Comparing the HMM and HSMM gated portfolios to each other, the differences are quite small. The HMM gated strategy has a slightly higher annualized return and Sharpe ratio than the HSMM gated strategy over the full sample. The HSMM gated strategy, on the other hand, exhibits a somewhat smaller maximum drawdown and slightly lower CVaR. The bootstrap confirms that the Sharpe ratio difference between the two gated strategies is far from significant (p = 0.72).

We can also look at the performance of the sleeves conditional on being selected by each model. On the days when the HMM chooses the mean reversion sleeve, that sleeve delivers very high risk adjusted returns with a Sharpe ratio of 1.32, albeit with substantial volatility. On days when the HMM chooses the momentum sleeve, the returns are moderate, and drawdowns are significantly smaller. The HSMM shows a similar pattern, though it tends to allocate to the mean reversion sleeve more frequently and achieves slightly lower conditional returns in that sleeve compared to the HMM. These observations confirm that both models are indeed tilting towards mean reversion in turbulent regimes and towards momentum in more favourable regimes.

Comparing the standalone performance of the momentum and mean reversion sleeves in Exhibit 1 with their performance on the days when they are selected by the models reveals substantial improvements. In both the HMM and HSMM, the sleeves' conditional Sharpe ratios increase by 34 to 42 percent for momentum and 68 to 91 percent for mean reversion, which indicates that the gated strategies are generally effective at directing capital into each sleeve when conditions are favourable. Although the overall gated strategies do not materially reduce the portfolio's maximum drawdown, the momentum sleeve itself largely avoids extreme losses, with a maximum drawdown below thirty percent in each model because it is mostly switched off during crises. This pattern suggests that there is promise to enhance risk aversion further by designing a more conservative mean reversion sleeve.

Behaviour in Persistent Drawdowns

To test my hypothesis about duration dependence in persistent downturns, I identify prolonged drawdown episodes in SPY based on peak to trough declines and duration. I focus on periods where the market declined by more than 25 percent and stayed in drawdown for several months. Each episode runs from the market peak to the lowest close before SPY regains that peak, plus 40 trading days. The episodes I identified are the 2007-2009 financial crisis bear market, the Covid-19 crash and the more recent 2022 drawdown associated with rising interest rates and inflation concerns. Exhibit 8 provides a summary of these episodes.

Exhibit 8: Persistent Drawdowns in the SPY

ID	Peak date	Trough date	End date	Duration	Drawdown	Avg. Volatility
1	09/10/2007	09/03/2009	05/05/2009	396	-56.5%	32.6%
2	19/02/2020	23/03/2020	19/05/2020	64	-34.1%	52.9%
3	03/01/2022	12/10/2022	08/12/2022	236	-25.4%	24.0%

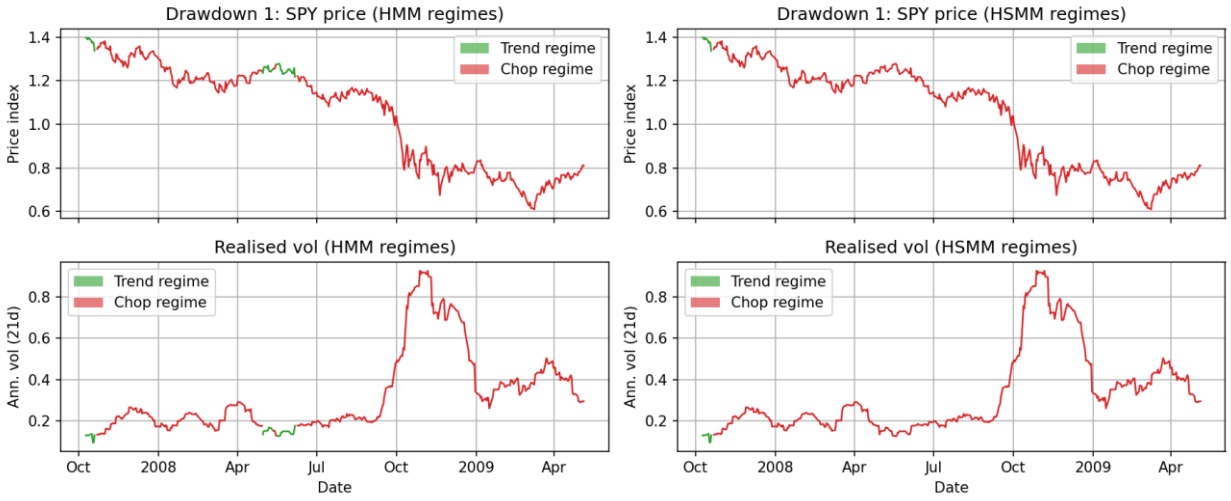
For each episode, I compute the cumulative return, maximum drawdown and daily CVaR of the HMM and HSMM gated portfolios over the episode. I also record how often each model switches regimes and the fraction of days it spends in the defensive mean reversion sleeve. The ID number matches the episode to Exhibit 8.

Exhibit 9: Model Performance in Persistent Drawdown

ID	Model	Return	Max Drawdown	CVaR 95%	Regime Switches	% Defensive
1	HMM	179.4%	-27.9%	6.0%	5	91.7%
	HSMM	167.5%	-27.9%	6.0%	1	97.7%
2	HMM	44.0%	-45.8%	9.6%	1	93.8%
	HSMM	44.0%	-45.8%	9.6%	1	93.8%
3	HMM	3.3%	-18.5%	3.3%	6	92.8%
	HSMM	8.6%	-18.4%	3.3%	1	99.6%

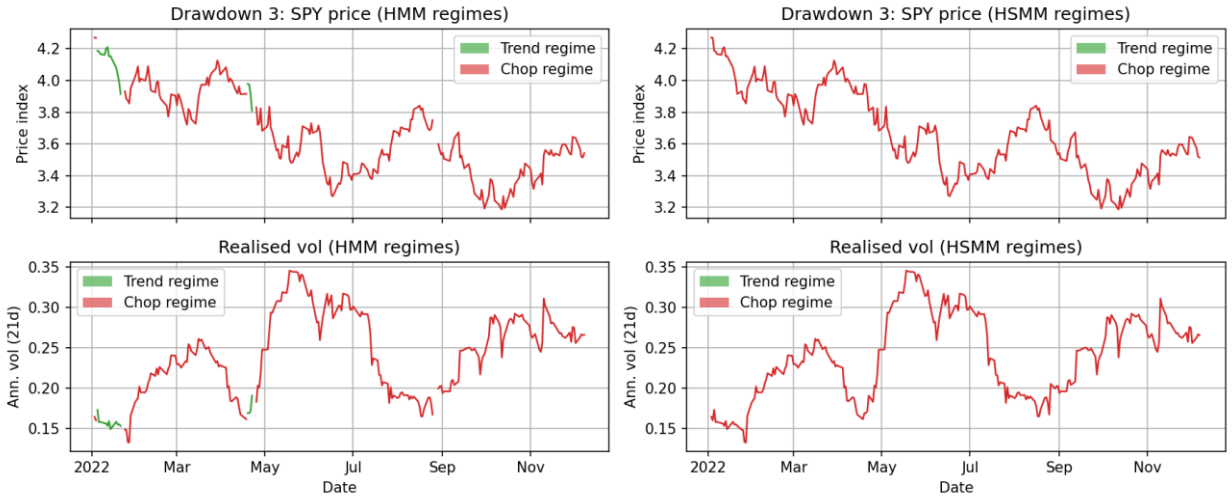
A visual comparison of the two models' regime predictions for the first persistent drawdown is shown in Exhibit 10.

Exhibit 10: HMM vs. HSMM allocation during the 2007-2009 financial crisis bear market



During the drawdown associated with the financial crisis (ID 1), both models are highly profitable with cumulative returns over 160% in just this period and this shows the strength of the mean reversion sleeve in handling drawdowns and high volatility markets. From a risk perspective, both models roughly halve the depth of the drawdown relative to the market, with the same maximum drawdown of 27.9 percent against a peak to trough drawdown of 56.5 percent in SPY. The HMM switches between regimes several times, returning to the momentum sleeve twice during the partial recovery in spring 2008, and stays in the mean reversion sleeve from June 2008 through the crash. The HSMM is more conservative. It switches once, near the start, and spends the rest of the episode in the mean reversion sleeve. This is an example of the qualitative expectation that a duration based model can stay defensive for longer in an extended decline. However, the performance consequences of this behavior are non-existent. In this particular episode, the HMM gated strategy achieved better returns and performs equal to the HSMM gated strategy in terms of risk metrics with the same max drawdown and CVaR. This suggests that in some persistent drawdowns, it is favourable to go risk on and trend follow when conditions are favourable which the HSMM does not allow for.

Exhibit 11: HMM vs. HSMM allocation during 2022 drawdown



In contrast, during the 2022 downturn (ID 3), the HSMM's conservative behavior pays off. Both models achieve smaller drawdowns of roughly 18.5 percent compared to 25.4 percent in SPY. The HMM allocates to mean reversion for most of the episode but occasionally switches into the momentum sleeve when conditions appear to improve. The HSMM however, stays in the mean reversion sleeve for the whole episode, switching back to momentum only on its last day, similarly to the 2007-2009 drawdown. Over this period, the HSMM gated strategy achieves a higher cumulative return than the HMM gated strategy (8.6 vs. 3.3 percent) while maintaining similar drawdown and CVaR.

During the Covid-19 crash (ID 2), the two models behave identically. Both switch into the mean reversion sleeve once, stay defensive for 94 percent of the episode and earn 44 percent. The episode also shows the risk of that sleeve: within the crash the gated portfolios fell 46 percent from peak to trough, more than the 34 percent drawdown in SPY, before recovering.

These results support the idea that an HSMM can produce more persistent defensive positioning in extended downturns, but they also show that the performance compared to the HMM depends on the possibility for profitable regime-switching in a persistent drawdown episode. Duration awareness helps in some persistent episodes and hurts in others.

Behaviour in Drawdown Spikes

I next consider short lived volatility spikes and corrections. I identify episodes where SPY experiences drawdowns over a shorter period, accompanied by a temporary rise in realised volatility (cannot overlap with persistent episodes). This aims to analyse the second part of my hypothesis that HSMM could have difficulty switching regimes and adapting in quick downturns.

Exhibit 12: Drawdown spikes in the SPY

ID	Peak date	Trough date	End date	Duration	Drawdown	Avg. Volatility
1	19/07/2007	15/08/2007	29/08/2007	30	-9.0%	19.7%
2	26/01/2018	08/02/2018	23/02/2018	20	-10.1%	19.6%
3	20/09/2018	29/10/2018	12/11/2018	38	-10.1%	14.9%
4	02/09/2020	21/09/2020	05/10/2020	23	-8.6%	21.3%
5	16/07/2024	05/08/2024	19/08/2024	25	-8.4%	15.9%

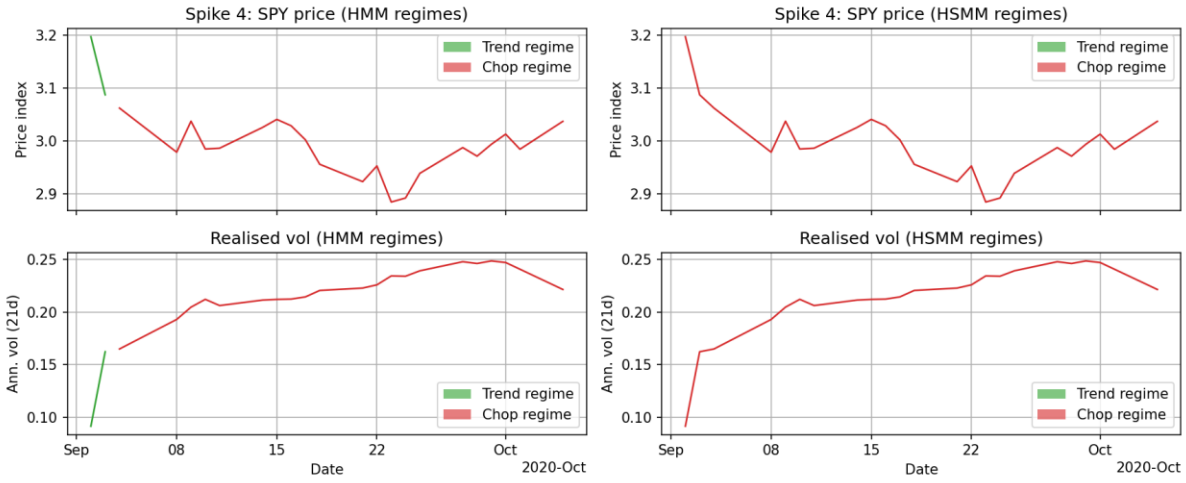
I compute the same metrics for each spike.

Exhibit 13: Model Performance in Drawdown spikes

ID	Model	Return	Max Drawdown	CVaR 95%	Regime Switches	% Defensive
1	HMM	4.9%	-11.4%	2.6%	1	77%
	HSMM	7.3%	-9.4%	2.6%	1	80%
2	HMM	-7.4%	-9.4%	3.8%	1	65%
	HSMM	-1.0%	-3.3%	2.4%	3	90%
3	HMM	-3.9%	-11.1%	3.7%	1	61%
	HSMM	-6.4%	-13.5%	3.7%	1	58%
4	HMM	-1.4%	-8.1%	3.5%	1	91%
	HSMM	3.5%	-3.8%	2.0%	0	100%
5	HMM	-10.3%	-12.3%	3.4%	1	40%
	HSMM	-10.3%	-12.3%	3.4%	1	40%

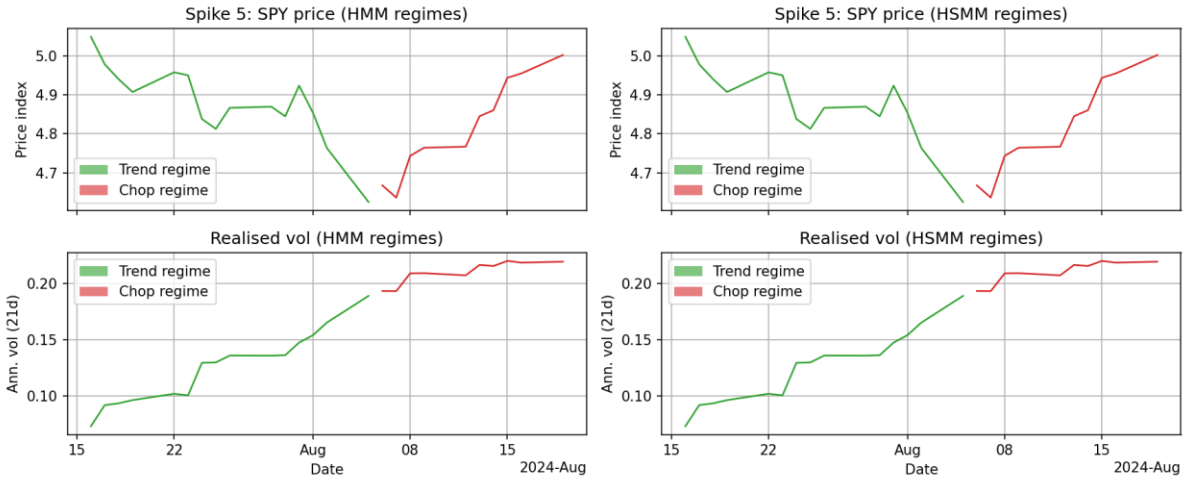
Exhibits 14 and 15 show the two models' regime predictions for the two most recent spikes.

Exhibit 14: HMM vs. HSMM allocation during the 2020 September spike



In the September 2020 spike (ID 4), the HSMM outperforms the HMM in all regards. The HMM model only slightly improves the max drawdown from 8.6 to 8.1 percent and achieves a loss of 1.4 percent. In contrast, the HSMM has a positive 3.5 percent return and a max drawdown of only 3.8 percent. While the HMM identifies a 'Chop' regime quickly, it is surprisingly the HSMM that is first to allocate to the mean reversion sleeve and benefit from this sleeves' performance. The substantial difference in performance with only minimally different behavior further shows the importance of quick adaptability for these models.

Exhibit 15: HMM vs. HSMM allocation during the 2024 July spike



In the July 2024 spike (ID 5), the behaviour of the two models is identical. They are slow to recognize the 'Chop' regime, only switching to the mean reversion sleeve after the drawdown has happened. This led to a negative 10 percent return for the models and a max drawdown of 12 percent.

Across the three earlier spikes (Exhibit 13), the HSMM does better in two (August 2007 and February 2018) and worse in one (October 2018). Together these episodes show the danger of quick drawdowns and the importance of quick adaptability in allocation models. I find that the

HSMM, contrary to my intuition, does not appear to be heavily weighed down by quick drawdown spikes and performs very similarly to the non-duration-aware model once again.

5. Discussion

This study set out to test whether an explicit duration component in a regime-switching model improves allocation between a momentum sleeve and a mean reversion sleeve. The central hypothesis was that a duration-aware HSMM would maintain defensive positioning longer in persistent high volatility drawdowns and thus reduce left tail losses relative to a standard HMM, while performing similarly in brief shocks.

The full sample results show that both regime gated strategies are very successful in improving risk adjusted performance relative to the underlying sleeves. Annualised returns of about 20 percent and Sharpe ratios near 0.9 are well above what either sleeve achieves on its own, although a block bootstrap places the improvement at the edge of statistical significance, so it is strong but not conclusive evidence. The regimes inferred by each model are intuitively interpretable and both models are able to avoid leaving the momentum sleeve fully exposed during major crises. However, the comparison between HMM and HSMM reveals only modest differences. The HMM gated strategy slightly outperforms the HSMM in terms of full sample return and Sharpe ratio, while the HSMM delivers a slightly smaller maximum drawdown and slightly lower daily CVaR. The duration aware model does not dominate in a clear and unconditional way. The gap is also small relative to estimation noise: re-estimating the HSMM with a different random initialisation of the EM algorithm changes the regime on 2.2 percent of days and moves its Sharpe ratio from 0.89 to 0.90, more than half the difference between the two models.

The behaviour of the regime processes helps to explain this outcome. The HMM produces two states that mainly differ in volatility. Average returns are similar across states, yet volatility is much higher in the chop state. The HSMM instead produces a more clearly “good” and “bad” state. Its trend state combines higher returns with low volatility, while the chop state has both lower returns and higher volatility. At the same time, HSMM regimes are shorter on average than HMM regimes. Run length distributions show that the HMM has heavier tails in this dataset, especially in the trend regime, while the HSMM alternates more frequently between states. In other words, the HSMM uses the duration structure to sharpen the separation between favourable and unfavourable regimes rather than to enforce very long trends.

The episode-based analysis illustrates how this difference in regime structure maps into allocation behaviour. In persistent drawdowns both models move strongly into the mean reversion sleeve and achieve high positive returns, which is largely driven by the strength of the defensive sleeve itself. The HSMM tends to stay defensive for the entire episode once it has switched. This behaviour is visible in the 2007–2009 crisis and the 2022 drawdown. In 2022, remaining fully defensive until the drawdown ended resulted in a higher cumulative return for the HSMM gated strategy while keeping drawdown and CVaR like the HMM. In contrast, during the financial crisis the HMM switches between sleeves a few times and ends up with a higher cumulative return for the same risk metrics. Here, the ability of the HMM to re-enter the momentum sleeve briefly during partial recoveries is rewarded.

Shorter volatility spikes tell a similar story. In the 2020 September correction, the HSMM again stays fully defensive and achieves a small positive return with much smaller drawdown and

CVaR than the HMM, which briefly moves into momentum and realises a mild loss. In the 2024 episode both models behave in the exact same way and performance is identical. These results suggest that the adaptability of the HSMM model is not significantly burdened by duration probabilities in quick drawdowns, contrary to my intuition.

Taken together, these findings partially support the hypothesis. The HSMM indeed produces more persistent defensive positioning in episodes that resemble extended downturns or spikes, and this can translate into better realised outcomes. At the same time, the benefits are conditional. When the underlying sleeves allow profitable tactical switching inside a prolonged drawdown, the more rigid duration structure of the HSMM can also lead to a missed upside.

These results are broadly consistent with those from Baitinger and Hoch (2024), who find that HSMMs fit the data better in sample but do not yield large and robust improvements in out-of-sample portfolio performance relative to simpler HMMs. The contribution of this project is to show that this conclusion extends to a different class of allocation problem. Instead of choosing static portfolio weights conditional on regimes, this study uses the regime model as a hard gate between two distinct sleeves with very different payoff profiles. The fact that the HSMM still only delivers incremental changes in risk measures suggests that duration modelling is not a straightforward improvement, even when it is cleanly isolated from other modelling choices.

For practitioners, a key implication is that the additional implementation effort of an HSMM does not appear to be justified if the goal is only to improve a two-sleeve switching rule of the type studied here. A well specified HMM already captures the main episodes of market stress and improves the conditional use of momentum and mean reversion. Duration modelling changes when the strategy spends time in the defensive sleeve and can alter the distribution of returns across episodes, yet it does not dramatically change full sample performance. In settings where interpretability, estimation stability and ease of calibration are important, a standard HMM remains an attractive benchmark and may be preferred.

At the same time, the HSMM's more persistent defensive behaviour in certain drawdowns highlights a useful design dimension. If an investor has strong preferences against exiting a defensive stance too early, for example due to regulatory capital constraints or behavioural risk aversion, imposing a duration structure that discourages rapid switching out of the chop state can be valuable even if the expected Sharpe ratio is not higher. In that sense, the HSMM can be seen less to extract extra alpha and more as a tool for encoding desired risk management behaviour in a systematic way.

6. Limitations and Future Research

Several limitations of this study should be noted. First, the analysis focuses on a single underlying asset, the S&P 500, and a single pair of sleeves. The conclusions therefore apply to this combination of index and strategies. Other asset classes, such as multi asset trend following, credit or commodities, may exhibit different regime structures and different interactions with duration modelling.

Second, the mean reversion sleeve used here is powerful but also aggressive. It trades daily across long and short positions in individual stocks, turning over about 91 percent of capital per day, and the results above ignore transaction costs, financing costs and constraints on shorting. Exhibit 16 quantifies the cost sensitivity by charging a cost per unit of capital traded. The mean

reversion sleeve breaks even at only 6 basis points, so on its own it would not survive realistic trading costs. The gated strategies are far more robust because they hold this sleeve only about a third of the time: the HMM-gated strategy breaks even at 22 basis points and the HSMM-gated strategy at 18. At 5 basis points their Sharpe ratios fall to 0.71 and 0.64. Costs therefore widen the gap between the two models, because the HSMM spends more days in the high-turnover sleeve. Since the gated strategies rely heavily on the strength of the mean reversion sleeve in crises, especially in persistent drawdowns, the absolute performance levels should be interpreted with caution. The mean reversion sleeve's return is also concentrated in time: 2008 and 2020 contribute about 60 percent of its summed daily returns, and excluding September 2008 to June 2009 lowers its Sharpe ratio from 0.69 to 0.57. Stock returns are CRSP price returns excluding dividends, which understates the long-only momentum sleeve by roughly the index dividend yield while leaving the long-short mean reversion sleeve largely unaffected.

Exhibit 16: Annual turnover, break-even cost and Sharpe ratio net of costs per unit traded

Strategy	Annual turnover	Break-even (bps)	Sharpe 0 bps	Sharpe 2 bps	Sharpe 5 bps	Sharpe 10 bps
Momentum	16x	61	0.47	0.46	0.43	0.40
Mean-reversion	230x	6	0.69	0.47	0.15	-0.39
HMM-gated	96x	22	0.91	0.83	0.71	0.50
HSMM-gated	115x	18	0.89	0.79	0.64	0.39

Third, the models are relatively simple. Both the HMM and HSMM use only two states and rely on a fixed set of five price-based features. More states could capture additional distinctions, such as early versus late crisis phases or low volatility drawdowns but would also increase estimation uncertainty. Similarly, the sojourn distributions in the HSMM are restricted to shifted Poisson forms that are re-estimated in an expanding window. Alternative specifications, such as Gamma or negative binomial durations or priors that encode stronger beliefs about minimum regime length, might produce different trade-offs between responsiveness and persistence.

Fourth, the allocation rule is intentionally simple. It uses hard gating into the sleeve linked to the most likely state at $t-1$ and ignores the magnitude of regime probabilities. Soft allocation rules that mix sleeves in proportion to regime probabilities, volatility targeting overlays and stop loss rules could all change how duration affects performance. This project is therefore best viewed as a clean laboratory where the duration component is isolated rather than as a fully optimised trading strategy.

These limitations suggest several directions for future research. One natural extension is to test HSMM gating in a multi asset setting with several sleeves that differ in both risk level and asset class. Another direction is to combine duration aware regimes with explicit risk control overlays, such as drawdown triggers that temporarily override the regime signal. A third avenue is to explore richer duration structures, for example models where the hazard rate of exiting a regime depends jointly on elapsed time and observable macroeconomic indicators. Finally, it would be valuable to conduct a broader Monte Carlo or bootstrap study to assess how sensitive the relative performance of HMM and HSMM is to sampling variation in crises and spikes, since the number of truly extreme episodes in historical data is limited.

7. Conclusion

This paper studied whether a duration aware hidden semi-Markov model can improve regime-based allocation between a momentum sleeve and a mean reversion sleeve relative to a standard hidden Markov model. Both models were constructed to be identical in all respects except for the treatment of regime durations. They used the same five SPY derived features, the same two-state structure, the same expanding window retraining schedule and the same hard gating rule that fully allocates to either momentum or mean reversion.

The empirical results show that regime gating itself is powerful. Both the HMM and HSMM gated portfolios substantially improve risk adjusted performance relative to the underlying sleeves. Sharpe ratios are close to 0.9 and the conditional Sharpe ratios of the sleeves on days when they are selected increase by 34 to 42 percent for momentum and 68 to 91 percent for mean reversion. At the same time, maximum drawdowns remain large, around fifty percent, which highlights the inherent risk of strategies that make concentrated bets on equity momentum and mean reversion, even when regimes are considered.

Comparing the two regime models, the HSMM delivers a more clearly separated pair of regimes with a high-return low-volatility trend state and a low-return high-volatility chop state. It also generates shorter regimes on average and more frequent switches. Although HSMMs are often motivated by their ability to capture more persistent regimes, in this application the estimated Poisson durations are on average shorter than the geometric durations implied by the HMM. The extra flexibility is therefore used mainly to sharpen regime separation rather than to lengthen regimes. In persistent drawdowns and volatility spikes, the HSMM tends to hold the mean reversion sleeve for longer than the HMM, which matches the qualitative idea of a duration aware model that stays defensive. This behaviour leads to better outcomes in some episodes and to foregone upside in others. Over the full sample the HSMM gated strategy achieves very similar performance to the HMM gated strategy, with slightly lower tail risk but also slightly lower return.

Overall, the evidence suggests that introducing explicit duration modelling in this setting does not transform the performance of the regime gated strategy. It changes when and where risk is taken, and it can be helpful in specific episodes, but it does not provide a large or robust improvement in out-of-sample metrics. Once transaction costs are considered, the comparison tilts further towards the HMM, which trades less. This finding is consistent with earlier work that compares HMM and HSMM based allocation rules and strengthens the conclusion that the main benefits of regime-switching arise from capturing broad differences between high and low risk environments rather than from fine tuning the distribution of regime lengths.

For investors and researchers, the main takeaway is that the choice between HMM and HSMM should be guided not only by statistical fit but also by the desired shape of defensive behaviour and by implementation costs. If one values simplicity and accepts some risk of exiting defensive regimes too early, a standard HMM is likely sufficient. If one cares more about encoding persistent caution once a crisis is identified, an HSMM can be a useful tool, even if expected returns are similar. Future work can explore these trade-offs in richer multi asset settings and under more realistic constraints, but the present study provides a first step in understanding how duration modelling interacts with regime-based switching between momentum and mean reversion sleeves.

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